ModelIt!

Student Activity Pack

Why Is Your Phone Designed to Be Addictive?

Modeling Dopamine Feedback Loops and Behavioral Design

Grade 11L1 | NGSS HS-LS1-3 | Cause and Effect

STUDENT NAME _____ DATE _____

ModelIt! — Discovery Collective

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1. **How many times do you think you check your phone in a day? What makes you reach for it even when you know nothing important has happened?**

2. **What do you think is happening in your brain when you see a notification pop up on your phone?**

3. **Why do you think scrolling through social media feels so different from reading a book — what makes one feel compulsive and the other feel like a choice?**

4. **Do you think phone addiction is about personal willpower, or is something else going on? Explain your reasoning.**

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Notification Frequency | Reward Variability | Dopamine Release | Tolerance Level | Compulsive Checking | Screen Time

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Social Media Default

Notifications: High | Reward variability: Maximum | 30-day simulation

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Digital Detox

Notifications: Off | 7-day simulation

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Ethical Redesign

Notifications: Moderate | Reward variability: Low (predictable)

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Social Comparison Index — The degree to which the platform encourages users to compare their lives to curated highlight reels of others, amplifying anxiety and driving return visits to monitor social standing

Algorithm Personalization — The sophistication of the AI content recommendation engine that learns individual preferences and delivers increasingly tailored content to maximize engagement duration

Sleep Disruption — The impact of blue light exposure and late-night dopamine stimulation on melatonin production and sleep quality, creating a secondary feedback loop where poor sleep increases vulnerability to compulsive behavior

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

What if the reason you can't put your phone down isn't about willpower — but about how your brain's own reward chemistry is being hijacked by design? Use evidence from your simulation to support your answers.

1. **How does your model demonstrate that phone addiction is an engineered outcome rather than a personal failure of self-control?**

2. **What does the tolerance mechanism in your model reveal about why screen time tends to increase over months and years rather than staying stable?**

3. **How could your model be used to design regulations that protect teens from exploitative app design without banning technology entirely?**

4. **What are the ethical implications of companies using neuroscience research to maximize engagement, knowing it creates addictive behavior?**

5. **What limitations does your model have in capturing the full complexity of human decision-making around technology use?**

8 STEM Challenge: Design an Ethical App Notification System

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YOUR ENGINEERING MISSION

Design a notification system for a social media app that maintains user engagement without exploiting dopamine feedback loops or creating addictive behavior patterns.

THE SCENARIO

A new social media startup wants to build a platform that is genuinely enjoyable but not addictive. They've hired your neuroscience consulting team to design a notification system that keeps users informed and engaged without triggering compulsive checking behavior. Your design must be based on your model's evidence about how different notification patterns affect dopamine, tolerance, and behavior.

Engineering Design Thinking

1. WHAT SPECIFIC NOTIFICATION FREQUENCY AND TIMING PATTERN DOES YOUR MODEL DATA SUGGEST WILL MINIMIZE COMPULSIVE CHECKING?

2. HOW WILL YOU MAKE REWARDS MORE PREDICTABLE WITHOUT MAKING THE PLATFORM BORING?

3. WHAT USER CONTROLS WILL YOU BUILD IN TO LET PEOPLE MANAGE THEIR OWN DOPAMINE EXPOSURE?

4. HOW WILL YOU MEASURE WHETHER YOUR DESIGN IS ACTUALLY LESS ADDICTIVE THAN CURRENT PLATFORMS?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Neuroethicists study the ethical implications of neuroscience and technology, working at universities, tech companies, and policy organizations to ensure brain science is used responsibly, earning $70,000–$130,000/year. UX Researchers who specialize in ethical design and digital wellness earn $85,000–$160,000/year at major tech companies, designing interfaces that respect users' cognitive health.